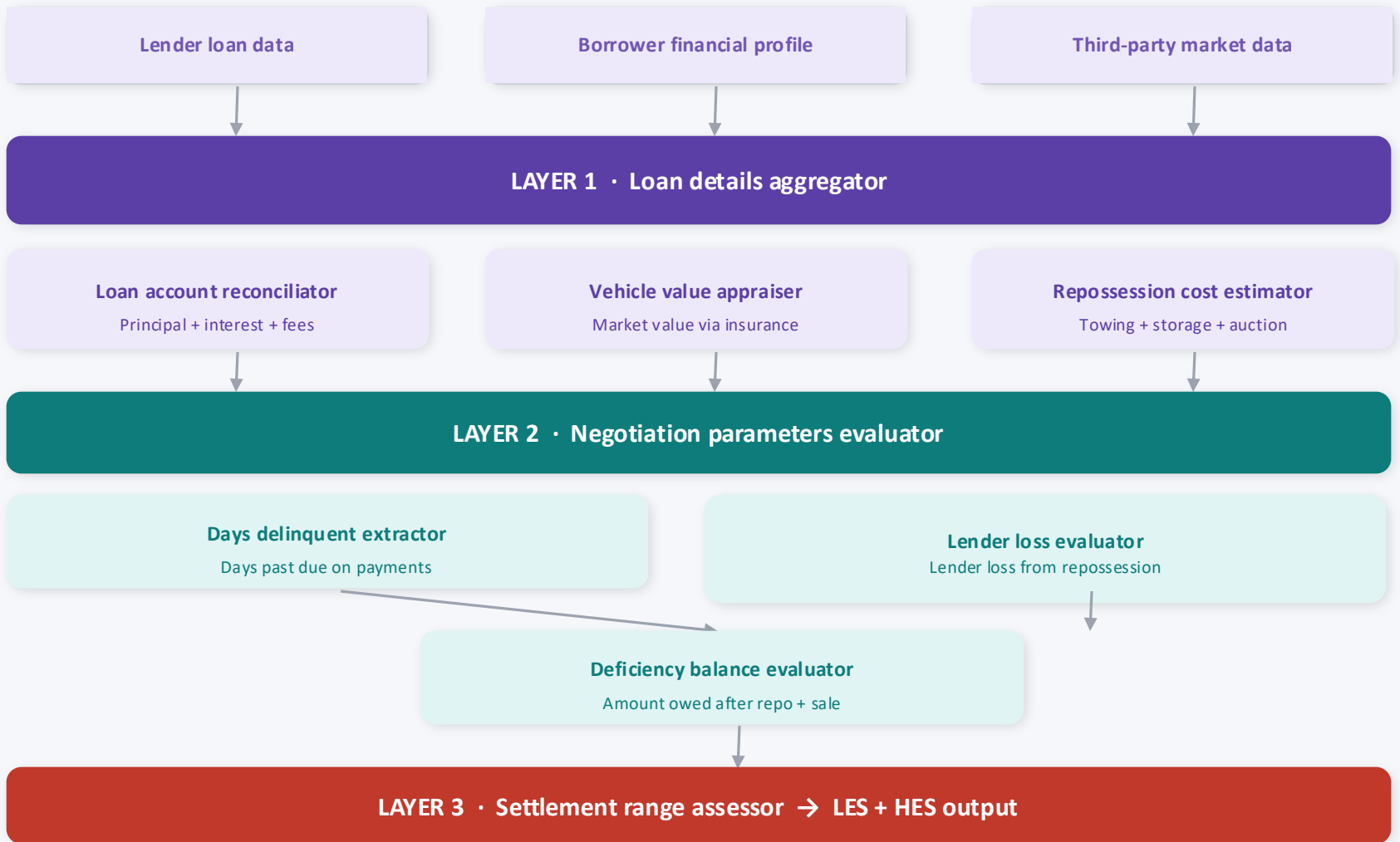


HIGH-LEVEL DESIGN

Auto Loan Settlement Negotiator

A layered architecture for computing realistic settlement ranges based on lender minimums and borrower repayment capacity.



Layer 1 ▪ Loan details aggregator

Aggregates all parameters required for settlement calculation. Three sub-processes feed raw data into the evaluation layer.

① Loan account reconciliator

- Total amount currently owed on the loan
- Includes outstanding principal balance
- Accrued interest to date
- Any assessed late fees or penalties
- Source: Lender loan management system

② Vehicle value appraiser

- Current fair market value of the vehicle
- Obtained from latest insurance valuation
- Or third-party sources (e.g. Kelley Blue Book)
- Used to determine lender's collateral recovery
- Directly impacts deficiency balance calculation

③ Repossession cost estimator

- Flat-rate estimate of repossession costs
- Towing and vehicle recovery fees
- Storage and lot fees during holding
- Auction processing and sale fees
- Lender uses this to assess net repo proceeds

All three sub-processes converge as inputs to Layer 2 — Negotiation Parameters Evaluator.

Layer 2 · Negotiation parameters evaluator

Evaluates lender risk exposure, borrower delinquency, and net financial position after a hypothetical repossession.

Days delinquent extractor

Extracts number of days loan payments are past due. Drives lender urgency score and settlement flexibility.

Lender loss evaluator

Evaluates total lender financial loss from repossession: vehicle shortfall vs. outstanding balance minus recovery costs.

Deficiency balance evaluator · Core anchor for settlement negotiation

Formula: $\text{Deficiency Balance} = \text{Total Amount Owed} - (\text{Vehicle Market Value} - \text{Repossession Costs})$
Represents the amount the borrower would still owe the lender AFTER repossession and vehicle sale.

Feeds into Layer 3 Settlement Range Assessor

Layer 3 · Settlement range assessor

Applies a rule set to compute the LES (floor) and HES (ceiling) of the negotiation corridor.

LES · Low-end settlement offer estimator

Floor of the negotiation range

Represents the minimum the lender is likely to accept.

Formula:

$LES \approx \text{Deficiency balance}$

- Repossession cost savings
- Lender urgency discount

Driven by:

- Days delinquent (higher = more flexible)
- Lender's estimated repo recovery

HES · High-end settlement offer estimator

Ceiling of the negotiation range

Represents the maximum a borrower would be expected to pay.

Formula:

$HES \approx \text{Total amount owed}$
(principal + interest + late fees)

Driven by:

- Borrower's lump-sum repayment capacity
- Specific lender settlement policy

vs

Negotiation corridor: LES ←-----→ HES

Output · Negotiated settlement value

The final value lands somewhere inside the negotiation corridor, shaped by three key variables:

01

Lender policy

Each lender has a specific minimum loss threshold. A lender with high delinquency volume may accept settlements closer to LES while conservative lenders hold near HES.

02

Borrower financial situation

Income, assets, other debts, and hardship documentation all influence willingness to settle. More documented hardship = lower final offer accepted.

03

Lump-sum repayment capacity

The ability to pay in full at settlement is a powerful lever. Lenders strongly prefer a clean lump-sum over a payment plan, often discounting heavily to close quickly.

Final negotiated settlement = f(lender policy, borrower financials, lump-sum capacity) within [LES, HES]

Component reference

Component	Layer / Type	Responsibility
Loan details aggregator	Layer 1 – Process	Aggregates all parameters needed for settlement calculation
Loan account reconciliator	Layer 1 – Sub-process	Determines total owed: principal + accrued interest + late fees
Vehicle value appraiser	Layer 1 – Sub-process	Gets current market value from insurance / third-party source
Repossession cost estimator	Layer 1 – Sub-process	Flat-rate estimate of towing, storage, and auction fees
Negotiation parameters evaluator	Layer 2 – Process	Evaluates lender risk exposure and borrower financial position
Days delinquent extractor	Layer 2 – Technical	Extracts days past due; drives lender urgency and flexibility
Lender loss evaluator	Layer 2 – Technical	Evaluates lender's net loss if repossession occurs
Deficiency balance evaluator	Layer 2 – Technical	Amount borrower still owes after repossession and vehicle sale
Settlement range assessor	Layer 3 – Process	Applies rule set to produce LES and HES estimate band
LES offer estimator	Layer 3 – Technical	Calculates lower bound: deficiency minus repo cost savings
HES offer estimator	Layer 3 – Technical	Calculates upper bound: total amount owed on the loan